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T12 – AI&DS

SUB : ASD&D

EXPERIMENT – 2

AIM : To Containerize Application using Docker.

LO Mapped : LO4

LO4 : Containerize , orchestrate , and deploy applications using Docker , Docker Compose , Kubernetes and Cloud Infrastructure .

THEORY :

1. Introduction to Docker

Docker is an open-source platform used to develop, package, deploy, and run applications in a standardized environment called a container. It uses operating-system-level virtualization to isolate applications and their dependencies without requiring a complete virtual machine.

Docker solves the common problem of "it works on my machine" by packaging an application along with its required libraries, dependencies, and configuration. This allows the application to run consistently across different environments.

2. Containerization

Containerization is the process of packaging an application and all its required dependencies into a lightweight, portable container.

Containers share the host operating system's kernel, making them generally faster and more lightweight than traditional virtual machines.

Advantages of containerization include:

- Portability

- Fast startup time
- Lightweight resource usage
- Application isolation
- Easy deployment
- Consistent environments
- Scalability

3. Docker Image

A Docker image is a read-only, immutable template that contains everything required to run an application. It includes:

- Application source code
- Runtime environment
- Libraries
- Dependencies
- Environment configuration
- Instructions for starting the application

Images are commonly created using a Dockerfile. An image can be used to create multiple containers.

For this experiment, the Docker image was named:

aryank

4. Docker Container

A Docker container is a running or stopped instance of a Docker image. It provides an isolated environment where the application executes.

A container is created from an image using the docker run command. Multiple containers can be created from the same image, and each container can have its own name, network configuration, and runtime state.

In this experiment, the container was named:

aryank-container

and a background container was named:

aryank-background

5. Difference Between Docker Image and Container

Feature	**Docker Image**	**Docker Container**
Definition	Template used to create containers	Instance of an image
State	Immutable/read-only	Has a runtime state
Purpose	Stores application and dependencies	Runs the application
Creation	Built using a Dockerfile	Created using docker run
Example	aryank	aryank-container
Multiple Instances	One image can create many containers	Each container is an individual instance
Lifecycle	Exists until deleted	Can be started, stopped, and removed

OUTPUT :

```
C:\Users\Lab203\docker-app>docker run -it myubuntu /bin/bash
root@a700eba76fe1:/# mkdir docker-app
root@a700eba76fe1:/# cd docker-app
root@a700eba76fe1:/docker-app# apt update
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [117 kB]
Hit:3 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:4 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [117 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [173 kB]
Get:6 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [376 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [387 kB]
Error: Release file for http://archive.ubuntu.com/ubuntu/dists/resolute-updates/InRelease is not valid yet (invalid for another 1h 8min 16s). Updates for this repository will not be applied.
```

```
root@a700eba76fe1:/docker-app# apt install -y nano
Installing:
  nano

Suggested packages:
  hunspell

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 34
  Download size: 289 kB
  Space needed: 893 kB / 1822 GB available

Get:1 http://security.ubuntu.com/ubuntu resolute-security/main amd64 nano amd64 8.7.1-1ubuntu0.1 [289 kB]
Fetched 289 kB in 1s (281 kB/s)
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 79, <STDIN> line 1.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (Can't locate Term/ReadLine.pm in @INC (you may need to install the Term::ReadLine module) (@INC entries checked: /etc/perl /usr/local/lib/x86_64-linux-gnu/perl/5.40.1 /usr/local/share/perl/5.40.1 /usr/lib/x86_64-linux-gnu/perl5/5.40 /usr/share/perl5 /usr/lib/x86_64-linux-gnu/perl-base /usr/lib/x86_64-linux-gnu/perl/5.40 /usr/share/perl/5.40 /usr/local/lib/site_perl) at /usr/share/perl5/Debconf/FrontEnd/Readline.pm line 8, <STDIN> line 1.)
debconf: falling back to frontend: Teletype
Selecting previously unselected package nano.
(Reading database ... 26589 files and directories currently installed.)
Preparing to unpack .../nano_8.7.1-1ubuntu0.1_amd64.deb ...
Unpacking nano (8.7.1-1ubuntu0.1) ...
Setting up nano (8.7.1-1ubuntu0.1) ...
update-alternatives: using /bin/nano to provide /usr/bin/editor (editor) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/editor.1.gz because associated file /usr/share/man/man1/nano.1.gz (of link group editor) does not exist
update-alternatives: using /bin/nano to provide /usr/bin/pico (pico) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/pico.1.gz because associated file /usr/share/man/man1/nano.1.gz (of link group pico) does not exist
```

```
root@a700eba76fe1:/docker-app# nano app.py
root@a700eba76fe1:/docker-app# nano requirements.txt
root@a700eba76fe1:/docker-app# nano dockerfile
root@a700eba76fe1:/docker-app# ls
app.py  dockerfile  requirements.txt
root@a700eba76fe1:/docker-app# cat dockerfile
FROM python:3.12-slim

WORKDIR /app

COPY . .

CMD ["python", "app.py"]
root@a700eba76fe1:/docker-app# docker build -t my-python-app .
bash: docker: command not found
root@a700eba76fe1:/docker-app# exit
exit

What's next:
    Debug this container error with Gordon → docker ai "help me fix this container error"
```

```
C:\Users\Lab203\docker-app>docker cp
docker: 'docker cp' requires 2 arguments

Usage:  docker cp [OPTIONS] CONTAINER:SRC_PATH DEST_PATH|-
        docker cp [OPTIONS] SRC_PATH|- CONTAINER:DEST_PATH

Run 'docker cp --help' for more information

C:\Users\Lab203\docker-app>docker ps -a
CONTAINER ID   IMAGE         COMMAND                  CREATED          STATUS                          PORTS     NAMES
a700eba76fe1   myubuntu      "/bin/bash"              7 minutes ago    Exited (127) 2 minutes ago                friendly_rhodes
362f82d23c91   myubuntu      "/bin/bash"              20 minutes ago   Exited (0) 11 minutes ago                 nostalgic_snyder
6f79701bdf8a   ubuntu        "/bin/bash"              29 minutes ago   Exited (127) 23 minutes ago               peaceful_ishizaka
38b568816772   ubuntu        "/bin/bash"              56 minutes ago   Up 56 minutes                             cool_meitner
50afc5d495c1   ubuntu        "/bin/bash"              56 minutes ago   Exited (0) 56 minutes ago                 cranky_germain
c8d7d78863ff   nginx         "/docker-entrypoint.…"   19 hours ago     Created                                   mynginx
db1e61f5826e   ubuntu        "bash"                   19 hours ago     Exited (127) 19 hours ago                 friendly_buck
9252b211eddb   ubuntu        "/bin/bash"              20 hours ago     Exited (2) 20 hours ago                   hungry_dhawan
1ec237a42c13   ubuntu        "/bin/bash"              20 hours ago     Exited (0) 20 hours ago                   admiring_villani
b7d7f60bf28c   ubuntu        "/bin/bash"              20 hours ago     Exited (127) 20 hours ago                 suspicious_bhaskara
96d324b8c9b3   ubuntu        "/bin /bash"             20 hours ago     Created                                   ecstatic_ride
7d8611fb4d31   ubuntu        "/bin /bash"             20 hours ago     Created                                   interesting_dijkstra
a5f07e98484b   ubuntu        "/bin/bash"              20 hours ago     Exited (0) 20 hours ago                   sad_hofstadter
f8662871de4c   hello-world   "/hello"                 4 months ago     Exited (0) 4 months ago                   hungry_bohr
cbdae8d0d2f5   hello-world   "/hello"                 4 months ago     Exited (0) 4 months ago                   awesome_galileo

C:\Users\Lab203\docker-app>docker cp a700eba76fe1:/docker-app
docker: 'docker cp' requires 2 arguments

Usage:  docker cp [OPTIONS] CONTAINER:SRC_PATH DEST_PATH|-
        docker cp [OPTIONS] SRC_PATH|- CONTAINER:DEST_PATH

Run 'docker cp --help' for more information

C:\Users\Lab203\docker-app>docker cp a700eba76fe1:/docker-app C:\Users\Lab203
Successfully copied 4.1kB to C:\Users\Lab203
```

```
C:\Users\Lab203>cd docker-app

C:\Users\Lab203\docker-app>nano app.py
'nano' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Lab203\docker-app>notepad app.py

C:\Users\Lab203\docker-app>notepad requirements.txt

C:\Users\Lab203\docker-app>notepad dockerfile

C:\Users\Lab203\docker-app>dir
 Volume in drive C is Windows
 Volume Serial Number is 3EB9-B88F

 Directory of C:\Users\Lab203\docker-app

21-07-2026  05:45    <DIR>          .
21-07-2026  05:38    <DIR>          ..
21-07-2026  05:54                29 app.py
21-07-2026  05:45                73 dockerfile
21-07-2026  05:43                 0 requirements.txt
               3 File(s)            102 bytes
               2 Dir(s)  742,369,075,200 bytes free
```

```
C:\Users\Lab203\docker-app>cd\

C:\>cd users

C:\Users>cd lab203

C:\Users\Lab203>dir
 Volume in drive C is Windows
 Volume Serial Number is 3EB9-B88F

 Directory of C:\Users\Lab203
```

```
C:\Users\Lab203>cd docker-app

C:\Users\Lab203\docker-app>nano app.py
'nano' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Lab203\docker-app>notepad app.py

C:\Users\Lab203\docker-app>notepad requirements.txt

C:\Users\Lab203\docker-app>notepad dockerfile

C:\Users\Lab203\docker-app>dir
 Volume in drive C is Windows
 Volume Serial Number is 3EB9-B88F

 Directory of C:\Users\Lab203\docker-app

21-07-2026  05:45    <DIR>          .
21-07-2026  05:38    <DIR>          ..
21-07-2026  05:54                29 app.py
21-07-2026  05:45                73 dockerfile
21-07-2026  05:43                 0 requirements.txt
               3 File(s)            102 bytes
               2 Dir(s)  742,369,075,200 bytes free
```

```
C:\Users\Lab203\docker-app>docker build -t my-python-app .
[+] Building 4.1s (8/8) FINISHED                                                
```

CONCLUSION :

The experiment successfully demonstrated the basic use of Docker for containerizing and running a Python Flask application. We created a Docker image using a Dockerfile, ran and managed containers, viewed logs, inspected containers, and practiced essential Docker commands for images, containers, and Docker Hub.